Chapter - 1

Early Blight of Potato

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Abstract

Early blight of potato, caused by the fungus *Alternaria solani*, poses a significant threat to potato cultivation across the globe, resulting in considerable yield reduction under optimal conditions for the pathogen. This disease manifests through distinctive symptoms such as dark, ring-like lesions on the leaves, often accompanied by yellowing, leading to early leaf drop and diminished tuber quality. The pathogen thrives in warm environments with high humidity and prolonged leaf wetness, which facilitate spore germination and infection. It persists on infected plant debris and alternative hosts, contributing to ongoing disease cycles. The plant-pathogen interaction is intricate, with *A. solani* producing various enzymes and toxins to invade potato tissues, while the resilience of host plants is often insufficient due to a lack of robust genetic resistance. Moreover, the pathogen's rapid evolution and resistance to fungicides complicate management efforts. To combat early blight effectively, IDM strategies are vital, combining cultural practices and resistant cultivars with selective fungicide applications and biological controls. This chapter primarily explores the pathogen, symptomatology, plant-pathogen interactions and strategies for disease management.

Keywords: *Alternaria solani*, early blight, potato, disease management

Introduction

Early blight, caused by the fungal pathogen *Alternaria solani* Sorauer, is a significant foliar disease affecting potato (*Solanum tuberosum* L). Characterized by dark brown to black lesions displaying concentric rings, this disease creates a target spot effect, typically manifesting first on older, senescing leaves. Initially described by Ellis and Martin in 1882 as *Macrosporium solani*, the first reference to the fungus as a parasite and its association with potato leaf blight was made by Galloway in 1891 in Australia. Chester (1892) further documented the disease in the United States, noting its occurrence on potatoes and other Solanaceae crops, including tomato and brinjal. He observed that early blight progressed more

slowly than late blight, caused by *Phytophthora infestans* (Mont.) de Bary that plants severely affected by early blight produced lower yields and smaller tubers. The nomenclature surrounding early blight was established by Jones in 1893, who suggested the term early blight to differentiate it from late blight. This distinction arises from the disease's tendency to attack early-maturing potato cultivars more severely than those that mature later, while late blight impacts medium and late-maturing varieties more significantly. Jones also reported that severe early blight epidemics could shorten the growing season due to premature vine death, resulting in smaller, immature tubers. He noted the prevalence of the disease on leaves damaged by insects, which provided entry points for the fungus and recommended control measures including two to three Bordeaux sprays and later planting (Galloway, 1893). During the late 19th century, a significant debate emerged regarding the pathogen responsible for early blight. Jones (1895) confirmed that *Macrosporium solani* was pathogenic, capable of entering the host through insect wounds or stomata. However, another fungal species, belonging to the *Alternaria* genus, was also isolated from lesions but failed to produce characteristic early blight lesions in artificial inoculations. This species was implicated in causing "tip-burn" of potato leaves, leading to confusion between *Macrosporium* and *Alternaria*. Ultimately, Jones and Grout (1897) isolated two species from potato leaves, designating one as the parasite causing early blight, which they proposed to name *Alternaria solani*, while the other, identified as *Alternaria fasciculata* (Cook and Ellis) L.R. Jones and Grout, is now recognized as *Alternaria alternata* (Fr: Fr.) Keissl. Potato stands as a vital food crop cultivated across the globe, playing a crucial role in nutrition and food security. It serves multiple purposes, including the production of starch, various food products such as potato chips and livestock feed (Ahmadizadeh and Felenji, 2011). Rich in carbohydrates, potatoes are integral to both table consumption and processed food industries. The tubers are composed of approximately 80% water and 20% dry matter, with over 75% of this dry matter consisting of starch, along with proteins, fiber and minimal quantities of fatty acids (Prokop and Albert, 2008). Furthermore, potatoes are a significant source of essential minerals, including potassium, phosphorus and magnesium, as well as vitamins B1, B3 and B6 (Camire *et al*., 2009). The presence of Vitamin C and various phenolic compounds contributes to their role as potent antioxidants (Brown, 2005). The potato's combination of high yield potential and substantial nutritional value positions it as the fourth most important food crop worldwide, following wheat, rice and maize (Bertorelli *et al*., 2020; Olivares *et al*., 2021).

Among potato producing countries, China holds the title of the largest producer and consumer, with an impressive production output of 93.43 million tons (Mt). Following, India with a production of 60.14 Mt, Ukraine 21.36 Mt, United states of America 19.29 Mt, Russia 19.37 Mt, Germany 11.60 Mt, Bangladesh 10.43 Mt and France 8.60 Mt as per 2023 estimates from the Food and Agriculture Organization (FAOSTAT, 2023). In India, the majority of potato production is concentrated in Uttar Pradesh, which yielded 15,892 thousand tons from an area of 622.50 thousand hectares. Other significant contributors include West Bengal, with 12,600 thousand tons produced over 447 thousand hectares and Bihar, which produced 9,125.80 thousand tons from 330 thousand hectares, according to the National Horticulture Board's estimates for 2021, NHB2021. Early blight, caused by the fungus *Alternaria solani*, significantly threatens potato cultivation by affecting both the foliage and tubers of the plants. This disease can lead to yield losses ranging from 5% to as much as 50%, depending on the severity of the infection and the management practices employed. The economic implications are considerable, as farmers may face not only reduced yields but also increased costs associated with disease management and crop replacement. One of the challenges in controlling early blight lies in its polycyclic nature, allowing multiple epidemics to occur within a single growing season due to the pathogen's ability to produce substantial amounts of secondary inoculum. This rapid spread under favourable environmental conditions makes containment difficult, leading to repeated cycles of infection that exacerbate the overall impact on the crop. This chapter emphasises various aspects in relation to early blight of potato pathogen, symptomatology, epidemiology and disease development, plant-pathogen interactions, host resistance and challenges in disease management, integrated disease management strategies.

Causal Organism

The fungal pathogen *Alternaria solani* is classified under the domain Eukaryota, kingdom Fungi, phylum Deuteromycota, class Hyphomycetes, order Hyphales and series Porosporae. According to new classification in hierarchy, *Alternaria* fungi is classified in Kingdom: Fungi belonging to Phylum: Ascomycota; Subdivision: Pezizomycotina; Class: Dothideomycetes; Order: Pleosporales; Family: Pleosporaceae; Subfamily: mitosporic Pleosporaceae; Genus: *Alternaria*. Most of the *Alternaria* species are saprophytes that are commonly by soil or on decomposing host tissues (Bart and Thomma, 2003). There are about 299 species that are scheduled in

the genus *Alternaria* (Kirk *et al*., 2008). The colony morphology of *A. solani* shows considerable variation but is typically effuse, ranging in colour from greyish-brown to black, with a texture that can appear cottony, felt like or velvety. This fungus demonstrates rapid growth on various media, although sporulation requires specific conditions. Pigments.

Ranging from orange to dark red are often produced by the colony, imparting a characteristic coloration to the growth medium. The cells of *A. solani* are multinucleate, but the number of nuclei varies depending on the fungal structure. After nuclear division in hyphal cells, multiple septations occur, dividing the elongated tip cells into several multinucleate sections (King and Alexander, 1969). The conidiophores of *A. solani* are dark or olivaceous-brown, thick-walled and can be straight or slightly flexuous. These structures are septate and arise either singly or in small groups, measuring up to 110 µm in length and 6-10 µm in diameter (Ellis and Gibson, 1975). Conidiogenesis in *A. solani* is tretic, producing conidia that are typically pale to olivaceous-brown, either singly or occasionally in short chains. The conidia are obclavate to elongate, double-walled and measure 75-350 µm in length and 20-30 µm in diameter at their broadest part, with 0-8 longitudinal or oblique septa and 6-19 transverse septa (Rao, 1969). The beaks of the conidia are filiform, straight or slightly flexuous and septate, with a diameter of 5-9 µmand their length ranges from one-half to double the length of the conidium (Rao, 1969). Due to variability in spore dimensions, they may overlap with other large-spored *Alternaria* species, making identification challenging.

The genus *Alternaria* includes several species that infect a wide range of plants, particularly those in the Brassicaceae (cruciferous plants) and Solanaceae families. Notable species include

A. alternata, *A. brassicae*, *A. solani* and *A. brassicicola* (Ellis and Martin). Among these, *A. brassicae* and *A. brassicicola* are significant pathogens of oleiferous and vegetable Brassicas, respectively. While *A. brassicicola* primarily infects vegetable Brassicas, such as cabbages and broccoli, *A. brassicae* is more prevalent in oleiferous Brassicas like mustard and rapeseed. In the Solanaceae family, *A. solani* is a major pathogen, causing early blight in economically important crops such as potato and tomato. Additionally, *A. alternata* is known to cause *Alternaria* blight and other diseases in over 380 host plant species, highlighting its broad host range and agricultural significance. The genus *Alternaria* was first described by Nees in 1817 and is recognized for its cosmopolitan distribution. These pathogens are highly influenced by climatic conditions, with disease

incidence peaking during wet seasons and in regions with high precipitation (Hong and Fitt, 1995). The environmental conditions, particularly humidity and rainfall, create favourable circumstances for the proliferation and spread of *Alternaria* species, leading to severe outbreaks in susceptible crops. This adaptability to diverse climates and host plants makes *Alternaria* a persistent challenge for farmers worldwide. Effective management of these pathogens often requires an integrated approach, combining cultural practices, resistant crop varieties and timely fungicide applications to mitigate their impact on agricultural productivity.

Symptoms

Alternaria solani affects multiple parts of its host plants, including fruits, leaves and stems, at all stages of development, making it a destructive pathogen. The disease symptoms are characterized by the formation of spots on stems, leaves and fruits of the infected plants (Sherf and Macnab, 1986). Foliar Symptoms include the characteristic circular, dark-brown or black lesions with concentric ring patterns that appear on older, lower leaves. Initially, small brown spots surrounded by yellow discolorations appear on the lower leaves, which gradually enlarge into dark brown to black lesions with distinctive concentric rings. These lesions can result in severe defoliation and, in some cases, damping-off. The circular or oblong spots, which start as light brown to grayish or dark brown patches, range from less than 0.5 mm to 12 mm in diameter and move upward to younger leaves as the plant matures. This progression significantly reduces the plant's photosynthetic potential (Rotem, 1994). On the midribs of leaves, the lesions may coalesce into linear, sunken patches that weaken the host tissue. Over time, the development of prominent black spots on leaves and stems becomes evident, often leading to the blackening of entire leaves, the weakening of the stem and the formation of elongated lesions. Symptoms commonly begin on older leaves, which are more vulnerable due to their proximity to the ground and aging process. Despite its destructive impact, *A. solani* does not typically affect water and nutrient uptake, as it does not target the roots or vascular system. The pathogen survives as mycelium or spores on decomposing plant debris, as it lacks a sexual reproductive stage (Rotem, 1994). Weakened or stressed plant tissues, whether due to environmental conditions, wounding or other factors, are more susceptible to infection, demonstrating the opportunistic nature of *Alternaria* species. Once in proximity to weakened plant tissue, *A. solani* transitions to a highly parasitic lifestyle, making the distinction between its saprophytic and parasitic behaviour less apparent. This pathogen is a leading cause of

significant yield losses in crops of the Solanaceae family, such as potato and tomato, contributing to both economic and agricultural challenges. It not only causes declines in fresh produce but also affects stored products, further amplifying its economic impact. Furthermore,

A. solani often co-occurs with *A. alternata*, which can complicate disease management and exacerbate crop losses.



Fig: Early blight of potato, caused by *Alternaria solani* (Kemmitt, 2002)

Epidemiology and Disease Development of *A. solani*

The development and progression of *Alternaria solani* infection in host plants, are heavily influenced by environmental factors, particularly temperature and relative humidity. The germination of *Alternaria* spores occurs effectively across a broad temperature range, with optimal hyphal growth observed at 25-27°C (Peralta *et al*., 2005). *In vitro* studies have shown that sporulation of *A. solani* is temperature-dependent, with a broader sporulation spectrum between 24-27°C. At 27°C, spores become fully developed and visible within 13-14 hours. High air humidity, ranging from 95-100% for at least 9-18 hours, is critical for successful infection and disease progression (Rotem, 1994). These environmental conditions create a favourable setting for the pathogen to infect host plants and spread rapidly, leading to significant crop damage.

During infection, *A. solani* produces a variety of phytotoxins, including alternariol, altersolanol A, altertoxin, macrosporin and solanapyrone, which contribute to the development of destructive diseases in plants. These toxins, classified as host-specific toxins (HSTs), play a crucial role in determining the host range and enhancing the pathogen's virulence and pathogenicity (Parada *et al*., 2008). Additional toxins, such as tentoxin, brefeldin, AF-toxin, AK-toxin, AM-toxin, ACT-toxin, AS-toxin, maculosin and destruxin B, have also been associated with *Alternaria* species (Meena and Samal, 2019). *A. brassicae* produces four cyclic depsipeptide phytotoxins, including destruxin B, which is a significant contributor to its pathogenicity (Tewari

and Bains, 1997). These toxins not only damage plant tissues but also elevate the pathogen's ability to infect and cause severe disease in susceptible hosts.

Plant-Pathogen Interactions

Plants are constantly exposed to various environmental challenges due to their close interaction with their surroundings. Unfavourable conditions, such as water scarcity or overload, nutrient imbalances, extreme temperature fluctuations and insufficient or excessive light, can weaken plants and make them more susceptible to biotic stress factors, including bacterial and fungal pathogens (Valcu *et al*., 2009). These environmental stresses not only damage plants physically but also create conditions conducive to the development of diseases caused by biotrophic and necrotrophic pathogens. The interaction between plants and specific fungal pathogens is highly complex, as pathogens have evolved mechanisms to overcome plant defense, while plants have developed diverse strategies to counteract these threats. For instance, plants produce antifungal chemicals as part of their defense mechanisms, but pathogens often evade these defenses by adapting to the host environment (Grayer and Kokubun, 2001).

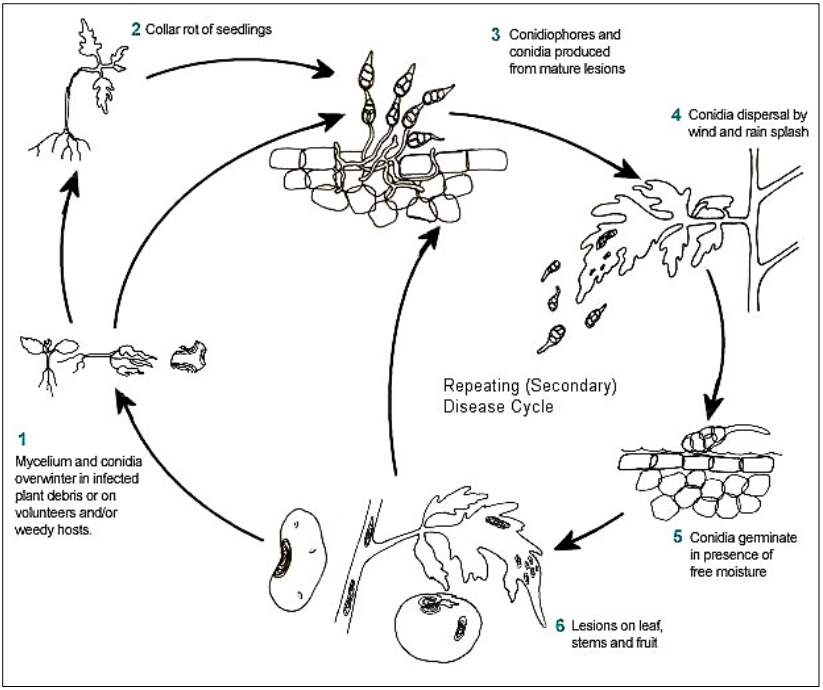


Fig: Life cycle of Early blight of potato (Kemmitt, 2002)

Pre-Infectional Defense Mechanisms in Plants

Plants possess multiple lines of defense to protect themselves from fungal pathogens. These include mechanical barriers, such as a thick cuticle layer and the production of pre-infectional metabolites, also known as phytoanticipins or prohibitins, which inhibit fungal spore germination and germ tube elongation (Arneson and Durbin, 1968). Members of the Solanaceae family, such as tomatoes, produce phytoanticipins like α-tomatine, a steroidal glycoalkaloid that plays a significant role in preventing fungal infections. Additionally, enzymes such as polyphenol oxidases, catalases and peroxidases, along with high deposits of epicuticular wax on leaves, contribute to reducing the incidence of infection by inhibiting conidial germination and germ tube formation. These pre-formed defense act as the first line of protection, but if they fail to restrict fungal penetration, the pathogen can invade the host plant and initiate infection.

Post-Infectional Defense Responses

When pre-formed defense are insufficient to prevent fungal infection, plants activate post-infectional defense mechanisms. These include the generation of reactive oxygen species (ROS), which act as alarm signals within the infected cell or neighbouring cells, triggering.

Structural fortification of the cell wall and hypersensitive responses, such as programmed cell death to limit pathogen spread (Glazebrook, 2005). Additionally, plants initiate systemic acquired resistance (SAR), which enhances their ability to resist subsequent infections and accumulate phytoalexins, a class of antifungal compounds that inhibit pathogen growth. These dynamic responses highlight the intricate interplay between plants and pathogens, where plants continuously adapt their defense strategies to counteract the evolving mechanisms of fungal pathogens. Despite these efforts, environmental factors such as temperature, humidity and light conditions significantly influence the success of fungal spore germination and infection, underscoring the importance of understanding plant-pathogen interactions for effective disease management.

Host Resistance and Challenges in Disease Management

Early blight affects potato, tomato, pepper and other members of the Solanaceae family. Developing host resistance is a critical component of integrated disease management for *A. solani*. However, resistance to this pathogen remains a significant challenge, as no highly resistant sources have been identified among the cultivated varieties of L. esculentum (Zhang *et al*., 2003). Many tomato cultivars are highly susceptible to early blight caused by

A. solani and limited information is available on the mechanisms of broad-spectrum or non-host resistance against this necrotrophic and toxigenic pathogen. The regular appearance of disease symptoms in various tomato varieties highlights the urgent need for research into resistance breeding and effective control strategies. Without robust resistance mechanisms, *A. solani* continues to cause substantial yield losses and economic damage, affecting both fresh and stored produce.

Integrated Disease Management (IDM)

The implementation of an Integrated Disease Management (IDM) strategy is a effective method for controlling this disease. This approach combines various techniques, including cultural practices, the use of resistant cultivars and the application of foliar fungicides (Warton and Kirk, 2012). Integrating these methods, growers can enhance their ability to manage disease outbreaks and minimize crop losses, ensuring healthier plants and improved yields.

Culture Practices

Early blight caused by *Alternaria solani* can be effectively managed through the adoption of various cultural practices. Crop rotation is a key strategy, with a 3-5-year rotation involving non-host crops, such as small grains, corn, or soybeans, helping to reduce the initial inoculum levels available for disease initiation (Agrios, 2005; Schultz and French, 2009; Warton and Kirk, 2012). Proper site selection, field sanitation and removal of infected plant debris through burning or ploughing under are also crucial for limiting pathogen persistence, as *A. solani* can overwinter in plant debris from one growing season to the next (Madden *et al*., 1978). Additional measures, such as providing balanced plant nutrition and ensuring adequate water management, are essential to reducing plant stress, which can make crops more vulnerable to infection. The use of certified, pathogen-free seeds and resistant varieties further minimizes the likelihood of disease outbreaks, while eradication of weed hosts in and around fields helps reduce the spread of inoculum. Post-harvest management, such as ploughing under all plant debris, storing lesion-free tubers in clean, dry, well-ventilated conditions at 40°F and avoiding tuber bruising during handling, is critical for reducing disease risks. Selecting well-drained, aerated fields, avoiding dense planting and facilitating proper air circulation also play a significant role in minimizing infection. Moreover, irrigation should be carefully managed avoiding overhead irrigation during cool, cloudy weather and ensuring plants dry before nightfall can significantly reduce environmental conditions favourable for fungal growth. Since

A. solani cannot infect tubers with intact periderm, storing tubers under conditions that promote rapid suberization is another effective measure to prevent post-harvest infections.

Use of Resistant Cultivars

Planting potato cultivars that exhibit lower susceptibility to early blight can help mitigate disease severity. However, research has indicated that host resistance does not significantly influence the initial onset of early blight (Shtienberg and Fry, 1990). While there are cultivars with good field resistance available, no commercial potato varieties or their wild relatives have shown complete immunity to early blight. Therefore, it is advisable to avoid planting highly susceptible cultivars in areas where early blight is common and disease pressure is high. Additionally, field resistance to foliar infection tends to correlate with plant maturity, with late-maturing cultivars generally exhibiting greater resistance compared to early-maturing varieties. Consequently, it is recommended not to plant early and late cultivars in the same or adjacent fields to reduce the risk of disease spread (Warton and Kirk, 2012).

Chemical Control

The use of fungicide spray programs is one of the most effective methods for managing early blight, particularly when applied from the early growing season until vine kill (Harrison *et al*., 1965; Douglas and Groskopp, 1974). During the initial stages of the disease, regular applications of contact fungicides are recommended to prevent infection. After flowering begins, 3-4 sprays of either systemic or contact fungicides should be applied, depending on the disease's progression. If symptoms appear before flowering, systemic fungicides should be used immediately. Effective fungicide application requires thorough coverage of the plant, especially the lower, senescing leaves, as these are the primary locations for early blight lesions to develop. Proper timing of fungicide applications can significantly reduce the frequency of sprays while maintaining yield levels (Van der Waals, 2002).

Systemic fungicides are essential during the flowering stage or when symptoms are visible, while protectant fungicides such as those containing chlorothalonil, macozeb should be applied at regular intervals starting at bloom. These applications can be scheduled every 7-10 days or adjusted based on weather conditions and the crop's developmental stage (Warton and Kirk, 2012). Early season applications, before the production of secondary inoculum, have minimal impact on disease spread. Therefore, fungicide sprays should begin at the first signs of disease or immediately after bloom to

achieve effective control. Alternating between contact and systemic fungicides is recommended to manage resistance and ensure effective disease control (Schultz and French, 2009). Spraying is generally more effective than dusting and applications should be made during calm weather, either in the early morning or evening, when wind speeds are below 10 mph for spraying or 5 mph for dusting. Dust formulations should contain at least 5-10% fungicide. Predictive models, such as scheduling the first fungicide application at 200 physiological days after crop emergence, can help optimize the timing of sprays (Warton and Kirk, 2012). Regular field inspections, particularly after plants reach a height of 12 inches, are also crucial for detecting early infections. Applying fungicides at the correct time and ensuring thorough coverage of foliage can effectively reduce the number of applications needed while minimizing disease spread and increasing crop yield.

Conclusion

Early blight of potato, caused by *Alternaria solani*, remains one of the most significant challenges in potato cultivation worldwide. This disease, characterized by dark, concentric lesions on leaves and tubers, can lead to severe yield losses if left unmanaged. Understanding the pathogen's biology, lifecycle and favourable environmental conditions is essential for crafting effective management strategies. By identifying the symptoms early and implementing integrated disease management practices, farmers can mitigate the potential damage caused by this pathogen. Integrated approaches, such as the use of resistant potato varieties, crop rotation, timely fungicide applications, and proper field hygiene, have proven to be effective in controlling early blight. Additionally, monitoring weather conditions and implementing predictive models can help farmers make informed decisions about disease prevention and treatment. Managing early blight requires a combination of scientific knowledge, practical application and ecological awareness. Continued research is crucial to develop innovative solutions, such as improved resistant cultivars or more effective biocontrol methods. By fostering collaboration between researchers, agricultural extension services and farmers, the fight against early blight can be more effective, ensuring the sustainability of potato production and food security for growing populations.

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